CONFIGURING A HUB-AND-SPOKE ARCHITECTURE USING AZURE FIREWALL

Project Overview

Project Title: Hub-and-Spoke Architecture using Azure Firewall.

Aim: To design and implement a secure hub-and-spoke network architecture in Microsoft Azure using Windows virtual machines in both Hub and Spoke resource groups. Azure Firewall is deployed in the Hub to control and inspect traffic between the Hub and Spoke Windows VMs.

Architecture Overview

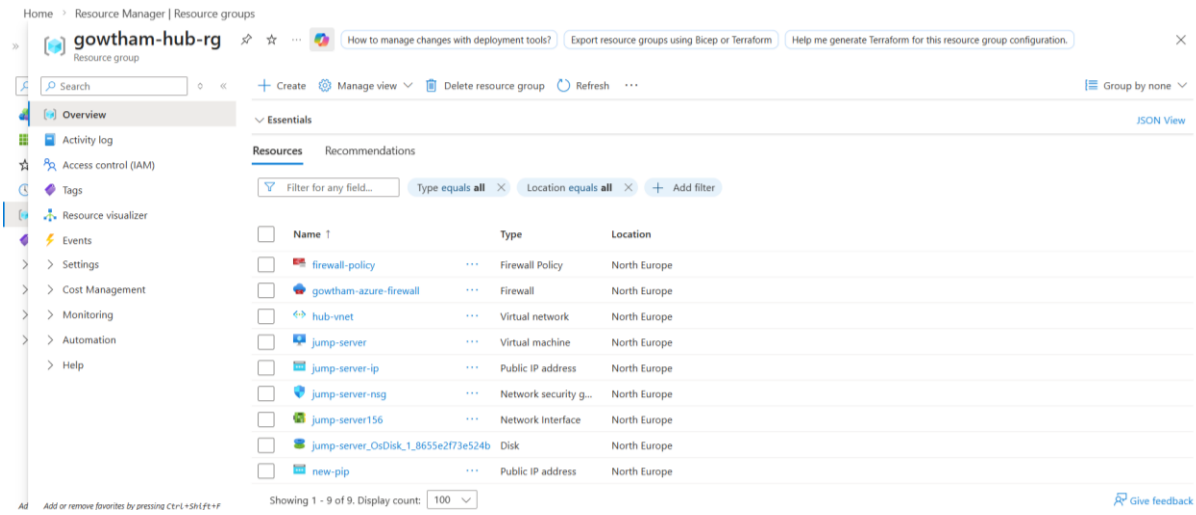
Windows VM (Hub Subnet) → Azure Firewall (Hub) → Windows VM (Spoke Subnet)

Azure Resources & Naming Convention

1. Hub Resource Group: gowtham-hub-rg
2. Spoke Resource Group: gowtham-spoke-rg
3. Hub Virtual Network: hub-vnet
4. Dedicated subnet for firewall:- AzureFirewallSubnet
5. Hub Subnet: hub-subnet
6. Hub Windows VM: jump-server
7. Spoke Virtual Network: spoke-vnet
8. Spoke Subnet: spoke-subnet
9. Spoke Windows VM: gowtham-spoke-vm
10. Azure Firewall: gowtham-azure-firewall

Step 1: Create Hub Resource Group

Create a resource group named **gowtham-hub-rg** to manage Hub resources such as the Hub VNet, Azure Firewall, and Hub Windows virtual machine.



Step 2: Create Hub VNet and Subnets

Create **hub-vnet**. Add AzureFirewallSubnet and hub-subnet

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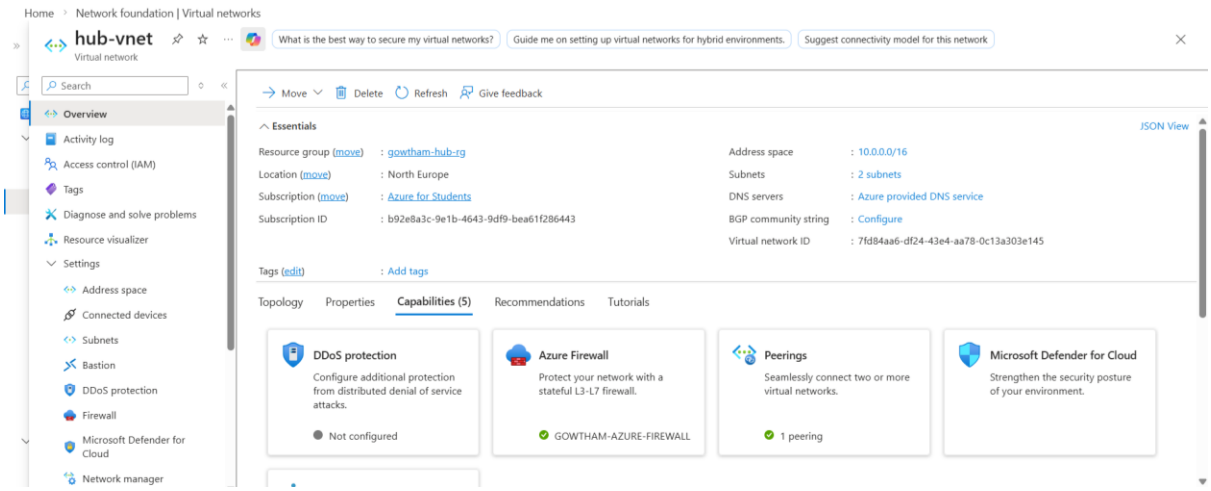
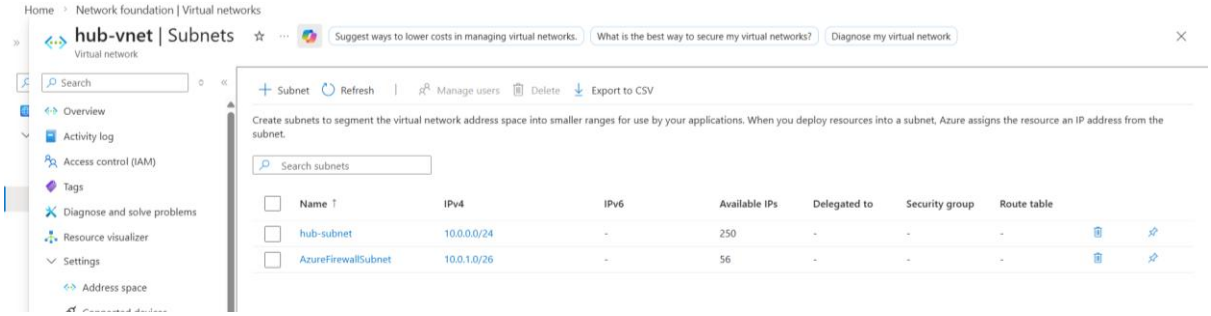


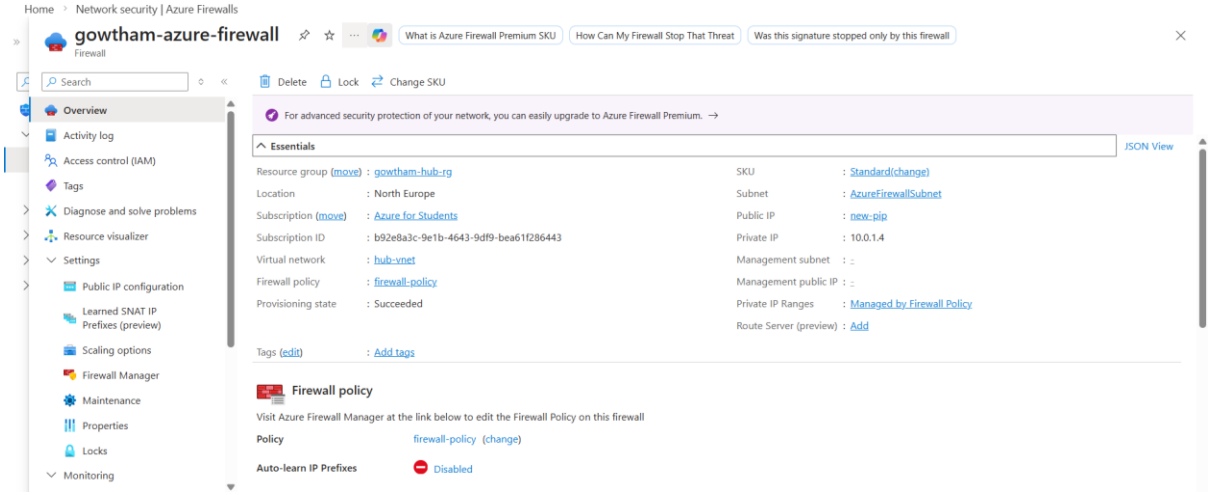
Hub-subnet



Step 3: Deploy Azure Firewall

Deploy Azure Firewall into the **AzureFirewallSubnet** and record its private IP address.

gowtham-azure-firewall



Step 4: Deploy Windows VM in Hub

Deploy a Windows virtual machine in **hub-subnet**. This VM acts as a jump server or management VM.

Jump-server

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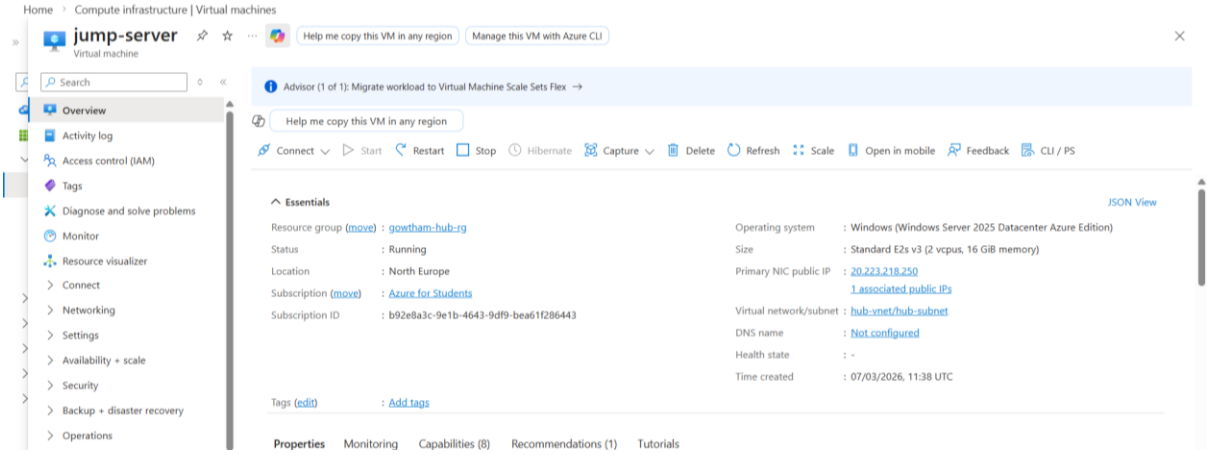
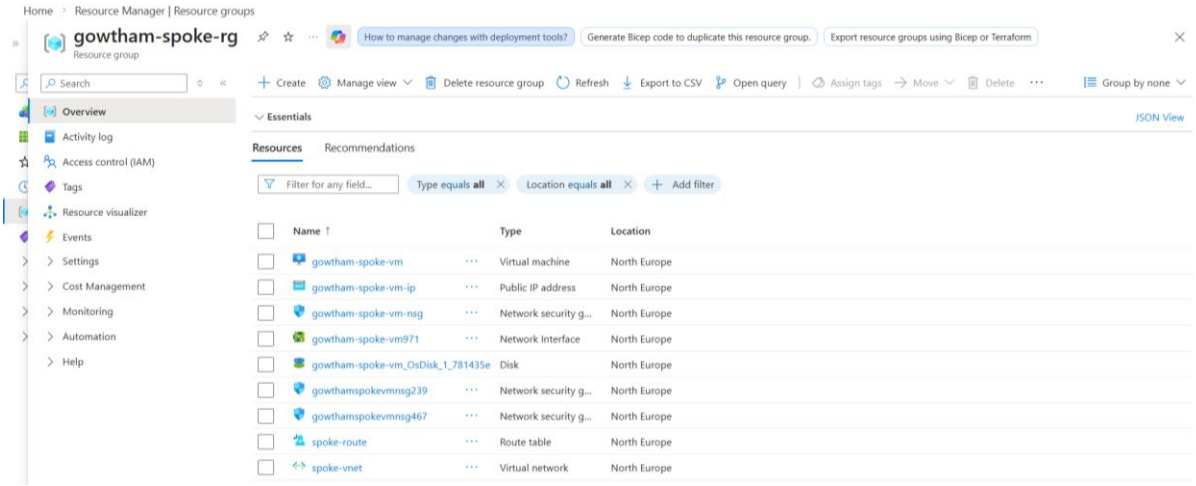


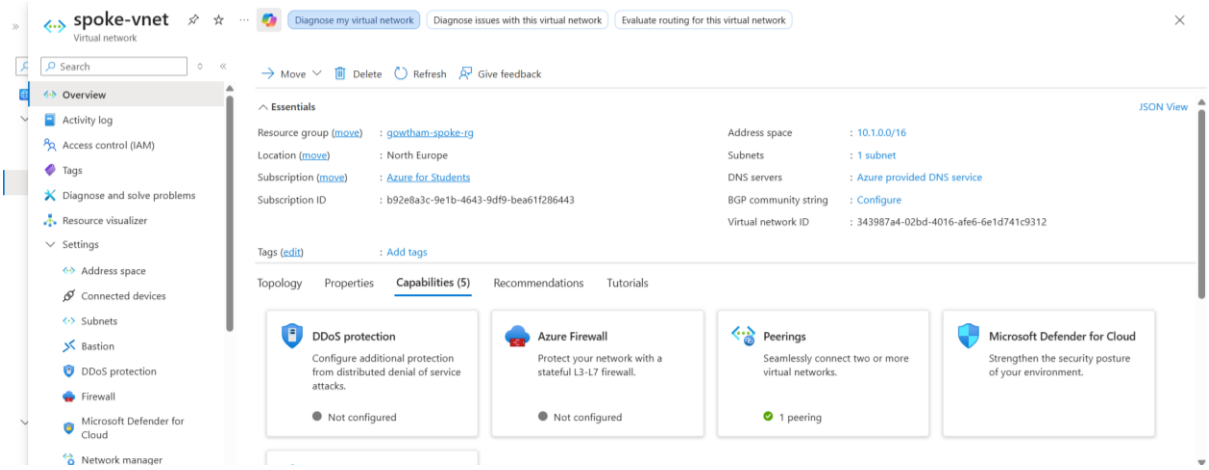
Step 5: Create Spoke Resource Group

Create a separate resource group named **gowtham-spoke-rg** to manage Spoke resources including the Windows workload virtual machine



Step 6: Create spoke VNet and Subnets

Create **spoke-vnet**. and spoke-subnet



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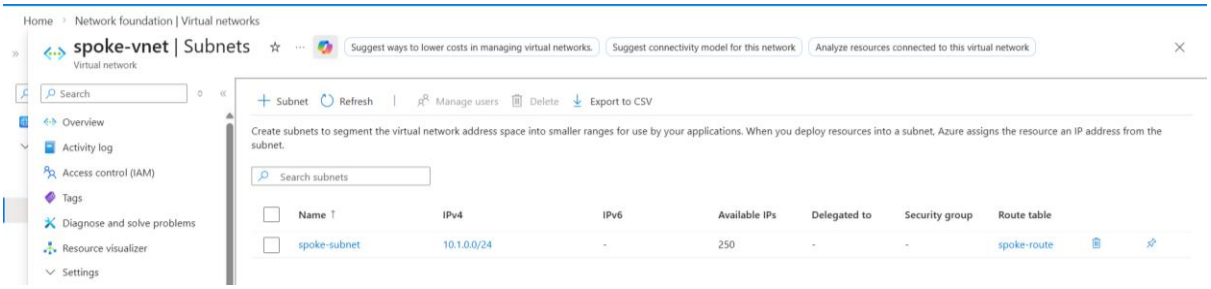
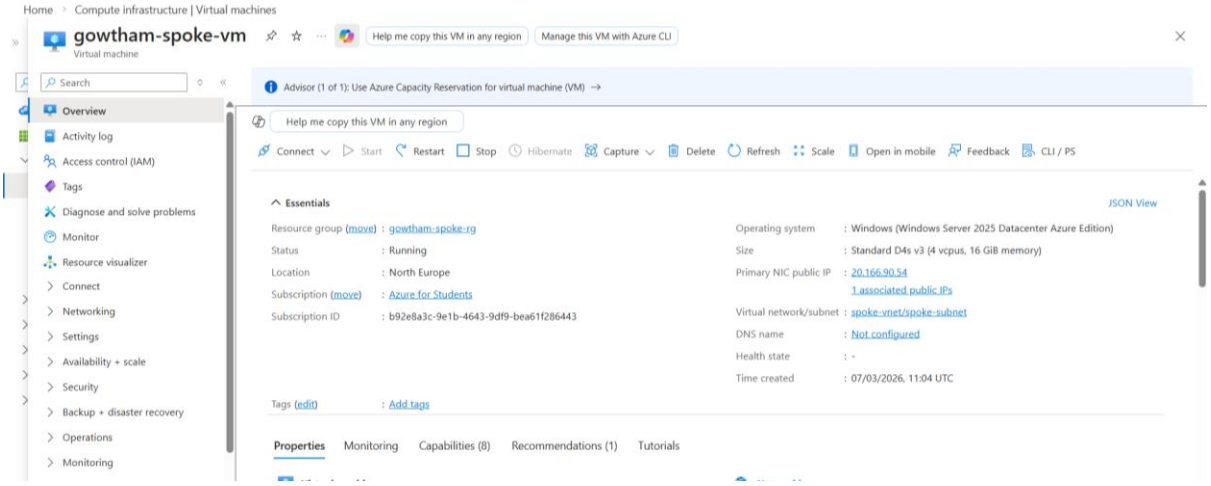
Spoke-subnet



Step 7: Deploy Windows VM in spoke

Deploy a Windows virtual machine in **spoke-subnet** named **gowtham-spoke-vm**



Step 8: Configure VNet Peering

Peer **spoke-vnet** with **hub-vnet**. Enable 'Allow forwarded traffic' and 'Use remote gateways' on the Spoke VNet.

Step 9: Configure Azure Firewall Rules

Create Network rules to allow RDP (3389) traffic from the Hub Windows VM to the Spoke Windows VM.

Step 10: Configure User Defined Routes (UDR)

Create a route table in the Spoke resource group and add a default route (0.0.0.0/0) pointing to the Azure Firewall private IP. Associate the route table with the Spoke Windows subnet.

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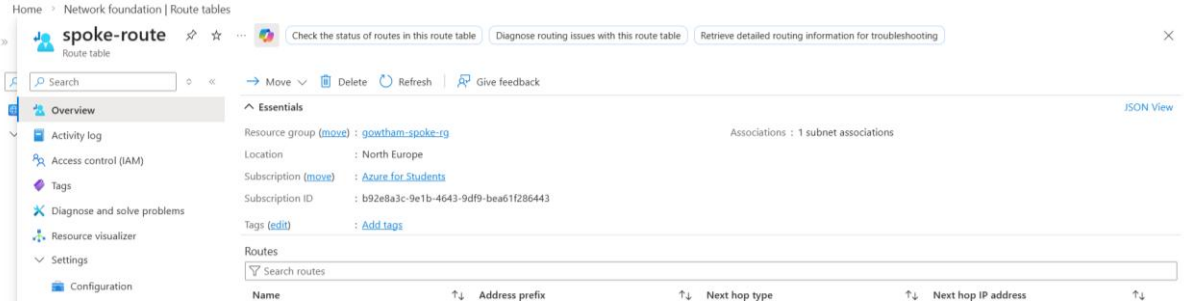
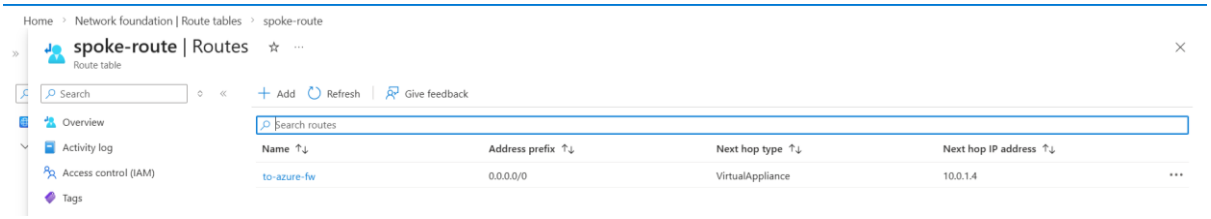
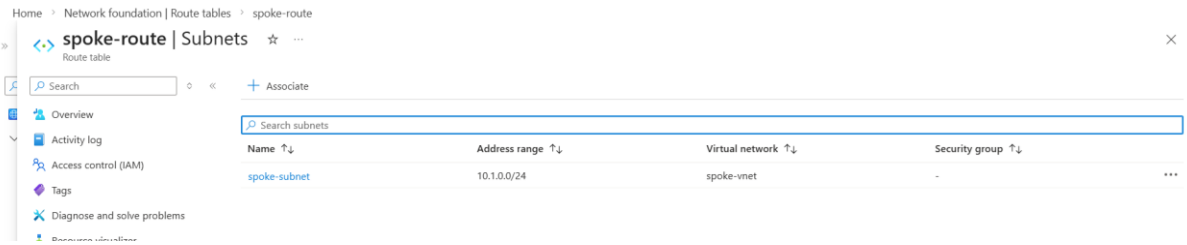
Create a route table in the Spoke resource group **spoke-route**



default route (0.0.0.0/0) pointing to the Azure Firewall private IP



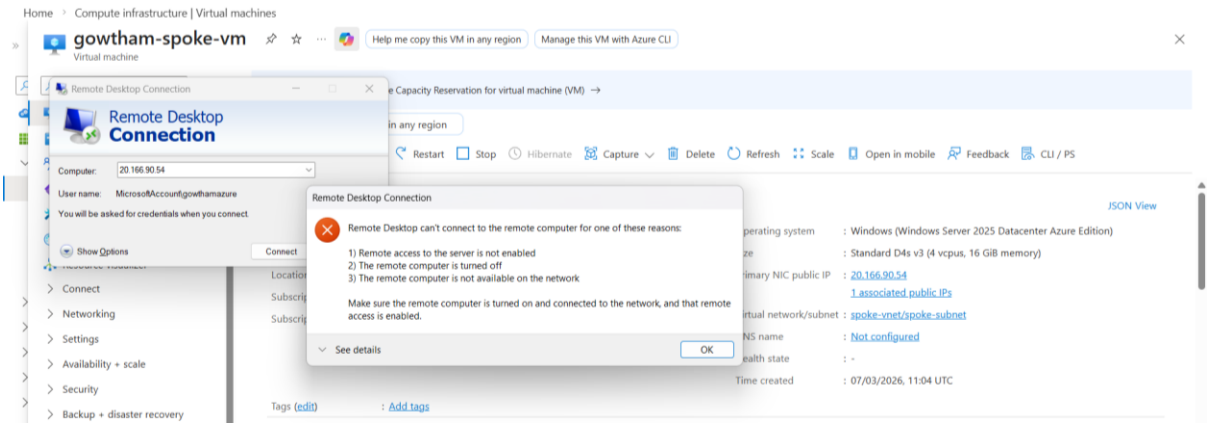
Associate the route table with the Spoke Windows subnet.



Validation & Testing

RDP into the Hub Windows VM and then RDP into the Spoke Windows VM. Verify that traffic flows through Azure Firewall and is logged.

gowtham-spoke-vm cannot connect because which is behind the firewall



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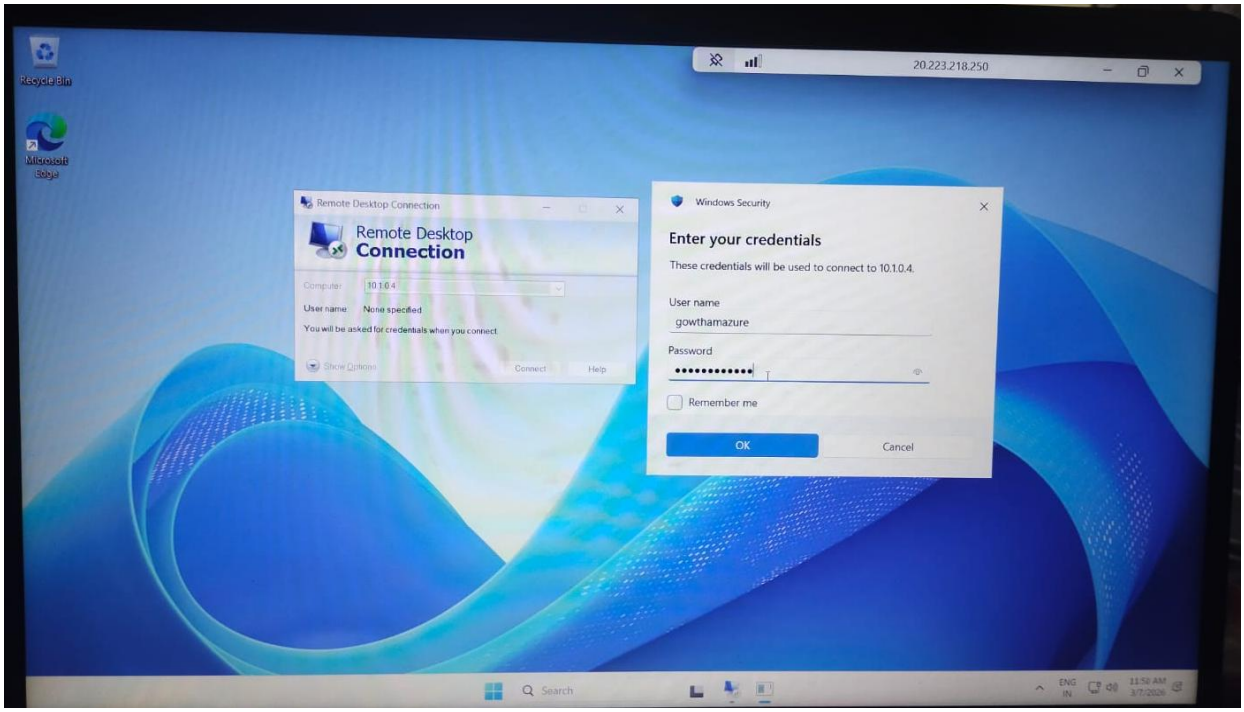
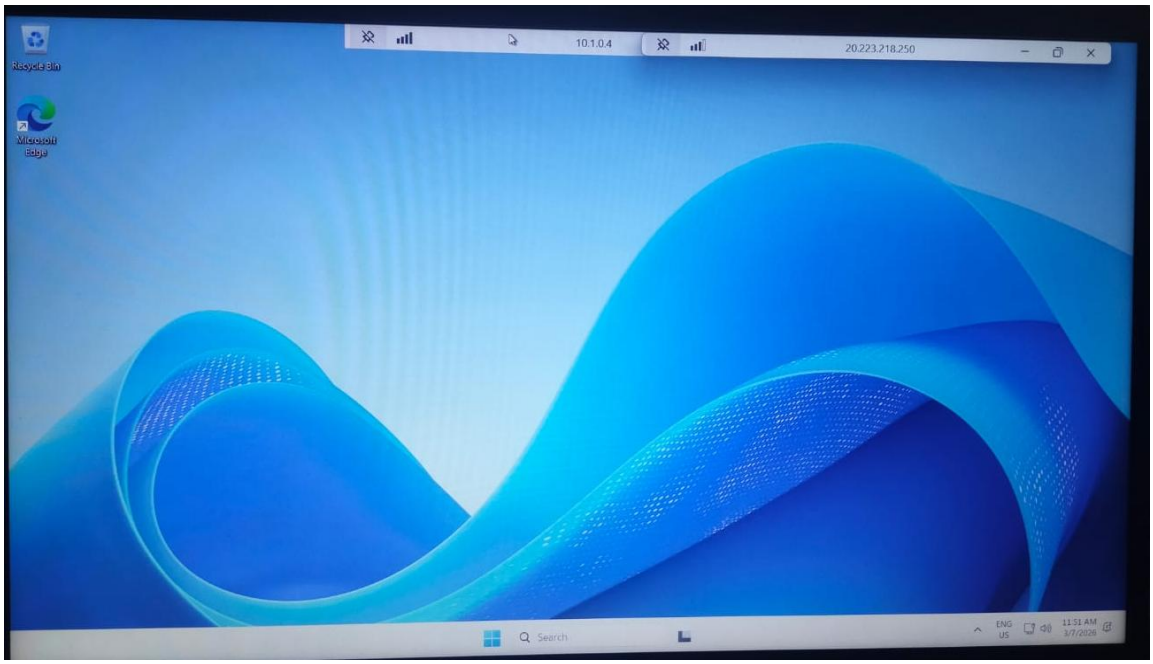
We can connect through jump-server into our gowtham-spoke-vm by using gowtham-spoke-vm private ip

First connect to jump-server using RDP from that connect to gowtham-spoke-vm using RDP



By using jump-server we can able to log-in into the gowtham-spoke-vm
By using gowtham-spoke-vm private-ip



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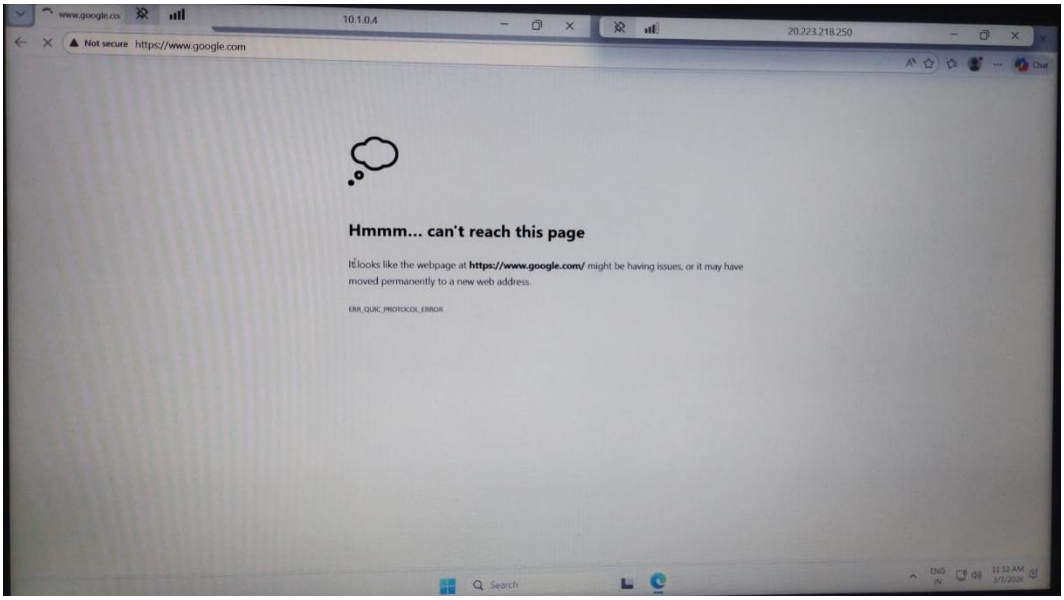
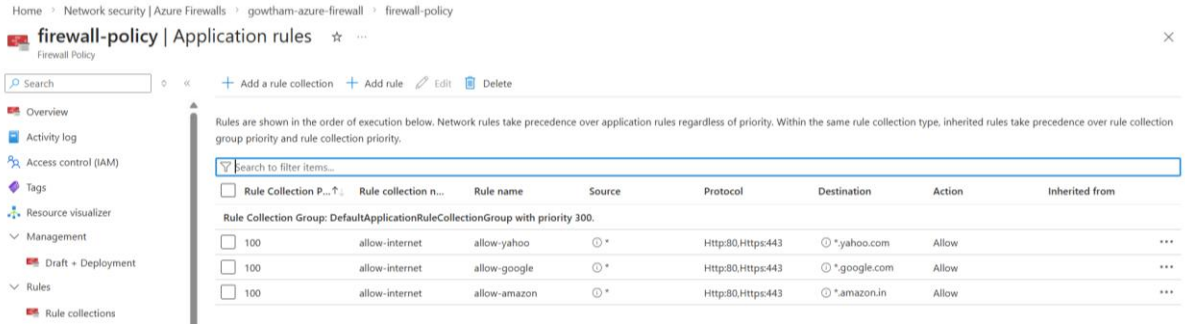
We connect through jump-server into our gowtham-spoke-vm by using gowtham-spoke-vm private ip But we cannot access any applications because our gowtham-spoke-vm is in behind the firewall.



After configuration of application rule(allow google) in firewall policy. We can access application.



We can access the application

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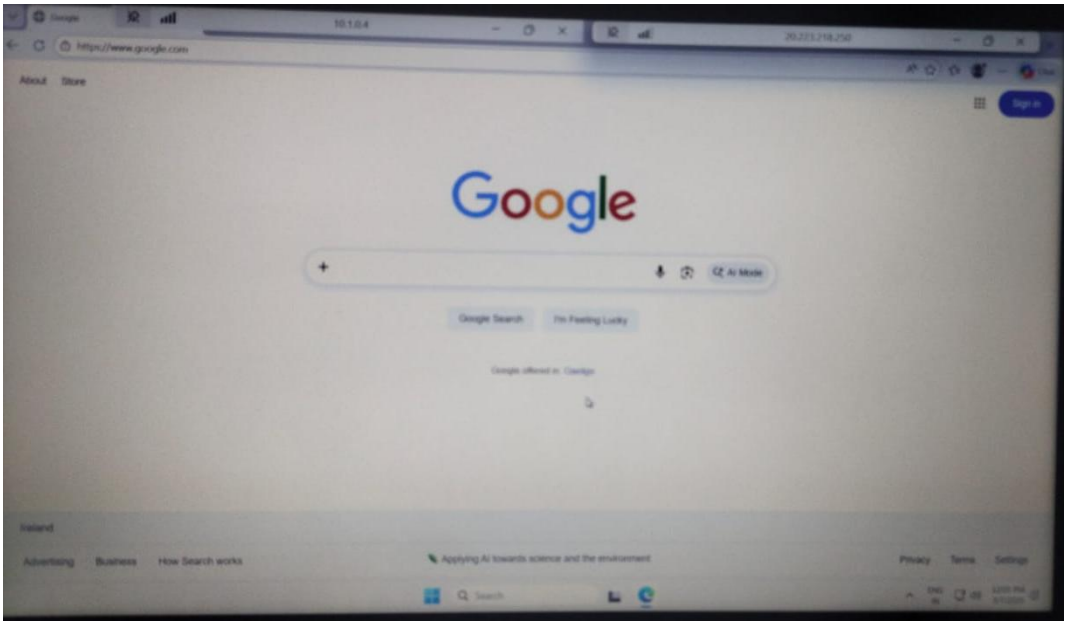

Conclusion

This implementation demonstrates a secure Windows-only hub-and-spoke architecture using Azure Firewall, separate resource groups, and controlled inter-VNet communication suitable for enterprise environments.

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